

On the Compactness of Palais-Smale Sequences for the p -Laplacian with Critical Nonlinearities in Smoothly Bounded Domains

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1 Introduction

In these notes we discuss some recent developments regarding the compactness of Palais-Smale sequences for the p -Laplacian in $\mathbb{R}^N$. In particular, we will introduce a new argument due to Meruci-Willem [3], which simplifies the approach taken by Struwe in [5] and [6] for the $p = 2$ case. Put informally, the key step in this new argument will be to focus on the dual space of $\mathcal{D}^{1,p}(\mathbb{R}^N)$, a homogeneous Sobolev space, rather than the “averaged decay” of the given Palais-Smale sequence. Despite this new perspective, the methodology behind his approach is well understood: first Meruci-Willem developed a non-existence result for the half-space and use this together with an iterative blow-up argument to decompose (after possibly passing to a subsequence) terms of the Palais-Smale sequence into non-trivial solutions of a limiting problem.

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Such compactness results are now indispensable, and highly sought after, tools for the analysis of partial differential equations, as they can be used to prove the existence of non-trivial solutions to various problems (e.g. blow-up solutions to Schrödinger equations and solutions to Yamabe equations – see Palatucci-Pisante [4] for more details). Despite their numerous applications, these compactness results are interesting in their own right. Fundamentally, they describe an asymptotic expansion of the Palais-Smale sequence in the energy space $\mathcal{D}^{1,p}(\mathbb{R}^N)$.

Henceforth, $N \geq 3$ is an integer, $\mu > 0$ is any real number, and Ω is a bounded domain in $\mathbb{R}^N$ having smooth boundary $\partial\Omega$. Unless stated otherwise, we will assume that $1 < p < N$. We denote by p' the conjugate Hölder exponent of p , and by p^* the Sobolev conjugate of p , i.e.

$$\frac{1}{p} + \frac{1}{p'} = 1 \quad \text{and} \quad p^* := \frac{Np}{N-p}. \quad (1.1)$$

Note that $p' \in (1, \infty)$ and that $p^* > p$. Finally, Δ_p stands for the p -Laplacian operator:

$$\Delta_p := \operatorname{div} (|\nabla u|^{p-2} \nabla u), \quad (1.2)$$

which we interpret in the weak sense.

Before proceeding, we will require some machinery from functional analysis and some simple preliminary results.

1.1 The Setup

In what follows, $\mathcal{X}$ and $\mathcal{Y}$ each denote normed vector spaces over the real numbers and $A : X \rightarrow Y$ denotes an operator, not necessarily linear. For the sake of convenience, $\mathcal{L}(\mathcal{X}, \mathcal{Y})$ is the real vector space of all bounded linear operators $\mathcal{X} \rightarrow \mathcal{Y}$. If $\mathcal{Y}$ is complete, then $\mathcal{L}(\mathcal{X}, \mathcal{Y})$ is a Banach space. Next, we reiterate the notion of a Fréchet derivative.

Definition 1. An operator $A : X \rightarrow Y$ is said to be *Fréchet differentiable* at a point $x \in \mathcal{X}$ if there exists a bounded linear operator $DA(x) : \mathcal{X} \rightarrow \mathcal{Y}$ such that

$$\lim_{\substack{h \rightarrow 0 \\ h \in \mathcal{X}}} \frac{\|A(x+h) - A(x) - DA(x)h\|_{\mathcal{Y}}}{\|h\|_{\mathcal{X}}} = 0.$$

The operator $DA(x)$ is called the *Fréchet derivative* of A at x . The *Banach valued* operator

$$DA : \mathcal{X} \rightarrow \mathcal{L}(\mathcal{X}, \mathcal{Y})$$

given by $x \mapsto DA(x)$ is called the *Fréchet derivative* of A , whenever it exists. If this operator exists, we say that A is *Fréchet differentiable* on $\mathcal{X}$. In the case that DA is continuous, we write $A \in C^1(\mathcal{X}, \mathcal{Y})$.

It is not hard to see that a Fréchet differentiable operator $\mathcal{X} \rightarrow \mathcal{Y}$ is automatically continuous. With this in mind, we are ready to formulate the setup of our main result. For N, p, μ and Ω as above, we are interested in $W_0^{1,p}(\Omega)$ -solutions to the problem

$$\begin{cases} -\Delta_p u + au^{p-1} \equiv \mu u^{p^*-1} & \text{in } \Omega, \\ u \geq 0 & \text{in } \Omega \end{cases} \quad (1.3)$$

where $a \in L^{N/p}(\mathbb{R}^N)$ is arbitrary, but fixed. Our main result will also involve weak solutions to the following *limiting problem*:

$$\begin{cases} -\Delta_p u \equiv \mu u^{p^*-1} & \text{in } \mathbb{R}^N, \\ u \geq 0 & \text{in } \mathbb{R}^N. \end{cases} \quad (1.4)$$

Next, let us consider the functional on $W_0^{1,p}(\Omega)$ given by:

$$\Phi : W_0^{1,p}(\Omega), \quad u \mapsto \int_{\Omega} \left(\frac{|\nabla u|^p}{p} + a \frac{|u|^p}{p} - \mu \frac{|u|^{p^*}}{p^*} \right). \quad (1.5)$$

By elementary techniques, it can be shown that Φ is Fréchet differentiable on $W_0^{1,p}(\Omega)$ (and hence continuous on this space) with derivative equal to

$$D\Phi(u)(h) = \int_{\Omega} \left(|\nabla u|^{p-2} \nabla u \cdot \nabla h + a |u|^{p-2} u h - \mu |u|^{p^*-2} u h \right), \quad (1.6)$$

for all $u, h \in W_0^{1,p}(\Omega)$. The function Φ is the energy functional associated with (1.3), and $D\Phi$ will be how we test to see whether or not a function $u \in W_0^{1,p}(\Omega)$ satisfies (1.3) in Ω . Broadly speaking, $u \in W_0^{1,p}(\Omega)$ will be called a *weak solution* to (1.3) if $D\Phi(u) = 0$ and $u \geq 0$ a.e. in Ω . To define the energy functional associated to (1.4), we will need to introduce the notion of a “homogeneous Sobolev space”, which we define below.

Definition 2. Given an open set $U \subseteq \mathbb{R}^N$, we define $\mathcal{D}^{1,p}(U)$ to be those functions $u \in L^{p^*}(U)$ having weak derivatives in $L^p(U)$. More precisely,

$$\mathcal{D}^{1,p}(U) := \{u \in L^{p^*}(U) : \nabla u \in L^p(\mathbb{R}^N; \mathbb{R}^N)\}.$$

We then equip $\mathcal{D}^{1,p}(U)$ with the natural seminorm

$$\|u\| := \|\nabla u\|_{L^p(U)}, \quad \forall u \in \mathcal{D}^{1,p}(U).$$

For general open sets U in $\mathbb{R}^N$, our definition makes $\mathcal{D}^{1,p}(U)$ into a topological vector space over $\mathbb{R}$. When $U = \mathbb{R}^N$, it is easy to see that $\|\cdot\|$ is actually a *norm* on $\mathcal{D}^{1,p}(U)$,

and thus $\mathcal{D}^{1,p}(\mathbb{R}^N)$ is a normed real vector space.¹ Finally, we denote by $\mathcal{D}_0^{1,p}(U)$ the closure of $C_c^\infty(U)$ in $\mathcal{D}^{1,p}(U)$. Note that $W^{1,p}(\mathbb{R}^N)$ is a *subset* of $\mathcal{D}^{1,p}(\mathbb{R}^N)$ by the Sobolev embedding theorem.

Remark 1.1. Typically one defines homogeneous Sobolev spaces in a different (and perhaps more natural) way. More explicitly, $\mathcal{D}^{1,p}(U)$ is usually defined to be the *completion* of

$$\{u \in L^{p^*}(U) : \nabla u \in L^p(U)\}$$

with respect to the norm $\|u\| := \|\nabla u\|_{L^p(U)}$. However, as previously mentioned, we are mostly interested in the dual of $\mathcal{D}^{1,p}(\mathbb{R}^N)$. Consequently, it does not matter which definition of $\mathcal{D}^{1,p}(\mathbb{R}^N)$ we work with. Indeed, let $\mathcal{X}$ be a normed space and denote by $\overline{\mathcal{X}}$ its completion. Since $\mathcal{X}$ is dense in $\overline{\mathcal{X}}$, any continuous linear functional $\mathcal{X} \rightarrow \mathbb{R}$ can be uniquely extended to a continuous linear functional on all of $\overline{\mathcal{X}}$. It then follows that $\mathcal{X}^* \cong \overline{\mathcal{X}}^*$ as Banach spaces.

We must be careful when working with both $W^{1,p}(\mathbb{R}^N)$ and $\mathcal{D}^{1,p}(\mathbb{R}^N)$. As mentioned above, it is true by the Sobolev embedding theorem that $W^{1,p}(\mathbb{R}^N)$ is a subset of $\mathcal{D}^{1,p}(\mathbb{R}^N)$. However, this does not make $W^{1,p}(\mathbb{R}^N)$ a subspace of $\mathcal{D}^{1,p}(\mathbb{R}^N)$. Certainly, the norm on $\mathcal{D}^{1,p}(\mathbb{R}^N)$ tests only the $L^p(\mathbb{R}^N; \mathbb{R}^N)$ -convergence of the gradients, and unfortunately the norm

$$\|u\| := \|\nabla u\|_{L^p(\mathbb{R}^N)}$$

is not equivalent to the $\|\cdot\|_{W^{1,p}(\mathbb{R}^N)}$ norm on $W^{1,p}(\mathbb{R}^N)$. Indeed, this is because the Friedrich-Poincaré inequality fails in $\mathbb{R}^N$. Hence, $W^{1,p}(\mathbb{R}^N)$ inherits a subspace topology from $\mathcal{D}^{1,p}(\mathbb{R}^N)$ different from its canonical topology. Despite this phenomenon, the smaller space $W_0^{1,p}(\Omega)$ can be embedded into $\mathcal{D}^{1,p}(\mathbb{R}^N)$. To see this, first note that every $u \in W_0^{1,p}(\Omega)$ can be viewed as an element of $W^{1,p}(\mathbb{R}^N) \subseteq \mathcal{D}^{1,p}(\mathbb{R}^N)$ by setting u to be identically 0 on Ω^c . By Friedrich's inequality, the norm $\|u\| = \|\nabla u\|_{L^p(\mathbb{R}^N)}$ is an equivalent norm on $W_0^{1,p}(\mathbb{R}^N)$ whence the observation follows.

When dealing with the limiting problem (1.4), we will be focused on weak solutions belonging to the space $\mathcal{D}^{1,p}(\mathbb{R}^N)$. To formalize this, we introduce (as above) an energy functional

$$\Phi_\infty : \mathcal{D}^{1,p}(\mathbb{R}^N) \rightarrow \mathbb{R}, \quad u \mapsto \int_{\mathbb{R}^N} \left(\frac{|\nabla u|^p}{p} - \mu \frac{u_+^{p^*}}{p^*} \right)$$

where $u_+ := \max(u, 0)$ denotes the positive part of $u \in \mathcal{D}^{1,p}(\mathbb{R}^N)$. This functional is

¹Note that we do not require $\mathcal{D}^{1,p}(\mathbb{R}^N)$ to be complete.

Fréchet differentiable on $\mathcal{D}^{1,p}(\mathbb{R}^N)$ with derivative

$$D\Phi_\infty(u)(h) = \int_{\mathbb{R}^N} \left(|\nabla u|^{p-2} \nabla u \cdot \nabla h - \mu u_+^{p^*-1} h \right),$$

for $u, h \in \mathcal{D}^{1,p}(\mathbb{R}^N)$. As before, we will say that $u \in \mathcal{D}^{1,p}(\mathbb{R}^N)$ is a weak solution to (1.4) if $D\Phi_\infty(u) = 0$ and $u \geq 0$ almost everywhere in $\mathbb{R}^N$.

To state our main result, we need only one last definition.

Definition 3. Let (u_n) be a sequence in $W_0^{1,p}(\Omega)$. We say that (u_n) is a *Palais-Smale* sequence (or simply a P.S.-sequence) for Φ if (u_n) is bounded, $\Phi(u_n)$ converges in $\mathbb{R}$, and

$$D\Phi(u_n) \rightarrow 0 \quad \text{in } W^{-1,p'}(\Omega)$$

as $n \rightarrow \infty$. Here, $W^{-1,p'}(\Omega)$ denotes the topological dual of $W_0^{1,p}(\Omega)$ over $\mathbb{R}$. Note that $W^{-1,p'}(\Omega)$ is a Banach space.

As it turns out, the requirement that (u_n) be bounded is redundant. It can be shown (see Struwe [5] for a proof in the $p = 2$ case) that any sequence (u_n) belonging to $W_0^{1,p}(\Omega)$ such that $\Phi(u_n)$ converges and $D\Phi(u_n) \rightarrow 0$ is necessarily bounded. However, since our goal is to illustrate the new method of proof used by Mercuri-Willem, it will not harm us to assume a priori that a P.S.-sequence is bounded.

Having given this critical definition, a coherent formulation of our main result (Theorem 2 in Mercuri-Willem [3]) is within reach.

Theorem 1 (Mercuri-Willem). *Assume that*

$$\inf_{\substack{u \in W_0^{1,p}(\Omega) \\ \|\nabla u\|_{L^p(\Omega)}}} \int_{\Omega} (|\nabla u|^p + a|u|^p) > 0 \quad (\star)$$

and let (u_n) be a Palais-Smale sequence for Φ such that

$$\|(u_n) - \|_{L^{p^*}(\Omega)} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

After possibly passing to a subsequence, there exists a weak solution $v_0 \in W_0^{1,p}(\Omega)$ to the problem (1.3) and a finite family $v_1, \dots, v_k$ in $\mathcal{D}^{1,p}(\Omega)$ of weak solutions to the limiting problem (1.4), with associated sequences $(y_n^i)_{n \in \mathbb{N}} \subset \Omega$ and $(\lambda_n^i)_{n \in \mathbb{N}} \subset \mathbb{R}_+$, such that for all

$i = 1, \dots, k$:

$$\frac{\text{dist}(y_n^i)}{\lambda_n^i} \rightarrow 0 \quad \text{as } n \rightarrow \infty, \quad (1.7)$$

$$\left\| u_n - v_0 - \sum_{i=1}^k (\lambda_n^i)^{\frac{p-N}{p}} v_i \left(\frac{\cdot - y_n^i}{\lambda_n^i} \right) \right\| \rightarrow 0 \quad \text{as } n \rightarrow \infty, \quad (1.8)$$

$$\|u_n\|^p \rightarrow \sum_{i=0}^k \|v_i\|^p \quad \text{as } n \rightarrow \infty, \quad (1.9)$$

$$\Phi(v_0) + \sum_{i=1}^k \Phi_\infty(v_i) = \lim_{n \rightarrow \infty} \Phi(u_n). \quad (1.10)$$

Furthermore, if $k > 0$, then each v_i is strictly positive in $\mathbb{R}^N$.

Although we will prove this result using the “new” argument given by Mercuri-Willem, there are several key steps that are *not* new. Typically, when proving a compactness result for Palais-Smale sequences, one can always expect to find the following standard technical ingredients used throughout the proof:

- a non-existence result for a limiting problem in the half-space $\mathbb{R}_+^N$;
- the use of a Lévy concentration function and a blowup argument;
- a lemma describing the limiting behaviour of sequences in $\mathcal{D}^{1,p}(\mathbb{R}^N)$ with respect to the limiting functional Φ_∞ and an analogous lemma for rescalings of sequences.

Properties regarding the particular concentration function used throughout our proof of Theorem 1 will be fully justified. However, we do not have the required tools to rigorously establish the non-existence result we require. Similarly, describing the limiting behaviour of a $\mathcal{D}^{1,p}(\mathbb{R}^N)$ sequence using the functional Φ_∞ is beyond the scope of these notes. Instead, we will use these technical results to give an elegant proof of Theorem 1.

2 Preliminary Results

As mentioned above, we begin by citing three results by Mercuri-Willem that will play a key role in the proof of Theorem 1. We should point out that these results are partial analogues of the $p = 2$ case, established by Michael Struwe in [6].

Theorem A. Any non-negative weak solution $u \in \mathcal{D}_0^{1,p}(\mathbb{R}_+^N)$ to the problem

$$-\Delta_p u \equiv \mu u^{p^*-1} \quad \text{in } \mathbb{R}_+^N$$

vanishes almost everywhere.

We shall also require the following technical result.

Theorem B. Suppose that $(\star)$ holds true and fix a Palais-Smale sequence (u_n) in $W_0^{1,p}(\Omega)$. In addition, assume that each of the following hold true:

- (1) u_n converges weakly to u in $W_0^{1,p}(\Omega)$,
- (2) $u_n \rightarrow u$ pointwise almost everywhere on Ω ,
- (3) $\|(u_n)_-\|_{L^{p^*}(\Omega)} \rightarrow 0$ as $n \rightarrow \infty$.

After passing to a subsequence, one has $\nabla u_n \rightarrow \nabla u$ pointwise almost everywhere on Ω and $\Phi'(u) = 0$. Furthermore, by defining $v_n := u_n - u$ we obtain a sequence in $W_0^{1,p}(\Omega)$ with the property that

- (1) $\lim_{n \rightarrow \infty} (\|u_n\|^p - \|v_n\|^p) = \|u\|^p$,
- (2) $\lim_{n \rightarrow \infty} \Phi_\infty(v_n) = \lim_{n \rightarrow \infty} \Phi(u_n) - \Phi(u)$,
- (3) $D\Phi_\infty(v_n) \rightarrow 0$ in $W^{-1,p'}(\Omega)$ as $n \rightarrow \infty$.

Finally, we cite a final result from Meruci-Willem which can be thought of as an analogue of the above for *rescalings* of the sequence (u_n) .

Theorem C. Assume $(\star)$ holds. Let (y_n) and (λ_n) be sequences in $\bar{\Omega}$ and $(0, \infty)$, respectively, such that

$$\frac{\text{dist}(y_n, \partial\Omega)}{\lambda_n^1} \xrightarrow{n \rightarrow \infty} \infty.$$

Suppose further that we have a sequence (u_n) in $\mathcal{D}^{1,p}(\mathbb{R}^N)$ such that the rescaled sequence

$$v_n(x) := \lambda_n^{\frac{N-p}{N}} u_n^1(\lambda_n x + y_n)$$

in $\mathcal{D}^{1,p}(\mathbb{R}^N)$ satisfies

- (1) $v_n \rightharpoonup v$ in $\mathcal{D}^{1,p}(\mathbb{R}^N)$ and pointwise almost everywhere on $\mathbb{R}^N$;
- (2) $\|(u_n)_-\|_{L^{p^*}(\Omega)} \rightarrow 0$ as $n \rightarrow \infty$;

(3) $\Phi_\infty(u_n) \rightarrow c$ and $D\Phi(u_n) \rightarrow 0$ in $W^{-1,p'}(\Omega)$ as $n \rightarrow \infty$.

Then $D\Phi_\infty(v) = 0$. Furthermore, after passing to a subsequence, we have

(1) $\lim_{n \rightarrow \infty} (\|u_n\|^p - \|w_n\|^p) = \|v\|^p$;

(2) $\Phi_\infty(w_n) \rightarrow c - \Phi_\infty(v)$ as $n \rightarrow \infty$; and

(3) $D\Phi_\infty(w_n) \rightarrow 0$ in $W^{-1,p'}(\Omega)$ as $n \rightarrow \infty$

with $w_n(z) := u_n(z) - \lambda_n^{\frac{p-N}{N}} v \left(\frac{z - y_n}{\lambda_n} \right)$.

Each of these results are taken directly from the paper due to Mercuri-Willem and their proofs can be found therein. In what follows, we establish the tools that we shall use freely within the proof of Theorem 1. The first lemma we prove is an elementary, yet infrequently stated, result regarding continuous functions.

Lemma 2.1. *Let (X, d) be a compact metric space and denote by $C(X)$ the real vector space of all continuous maps $X \rightarrow \mathbb{R}$, endowed with the norm*

$$\|\cdot\|_\infty := \sup_{x \in X} |\cdot(x)| < \infty.$$

If $\mathcal{F} \subseteq C(X)$ is equicontinuous and equibounded, then

$$\sup_{f \in \mathcal{F}} f \in C(X).$$

Namely, $\sup_{f \in \mathcal{F}} f$ is continuous on X .

Proof. First, we define a function $s : X \rightarrow \mathbb{R}$ via the rule

$$s(x) := \sup_{f \in \mathcal{F}} f(x);$$

since the family $\mathcal{F}$ is equibounded, this is clearly a well defined function. Next, consider any two points $x, y \in X$. Clearly,

$$\sup_{f \in \mathcal{F}} f(x) = \sup_{f \in \mathcal{F}} (f(x) - f(y) + f(y)) \leq \sup_{f \in \mathcal{F}} (f(x) - f(y)) + \sup_{f \in \mathcal{F}} f(y)$$

whence

$$\sup_{f \in \mathcal{F}} f(x) - \sup_{f \in \mathcal{F}} f(y) \leq \sup_{f \in \mathcal{F}} (f(x) - f(y)) \leq \sup_{f \in \mathcal{F}} |f(x) - f(y)|.$$

Similarly, we have

$$\sup_{f \in \mathcal{F}} f(y) - \sup_{f \in \mathcal{F}} f(x) \leq \sup_{f \in \mathcal{F}} (f(y) - f(x)) \leq \sup_{f \in \mathcal{F}} |f(x) - f(y)|.$$

Combining these two inequalities yields

$$\left| \sup_{f \in \mathcal{F}} f(x) - \sup_{f \in \mathcal{F}} f(y) \right| \leq \sup_{f \in \mathcal{F}} |f(x) - f(y)|.$$

Let now $\varepsilon > 0$ be given. By equicontinuity, there is $\delta > 0$ such that $|f(x) - f(y)| < \varepsilon$ for all $f \in \mathcal{F}$, whenever $d(x, y) < \delta$. For any such $x, y \in X$, we must therefore have

$$\left| \sup_{f \in \mathcal{F}} f(x) - \sup_{f \in \mathcal{F}} f(y) \right| \leq \sup_{f \in \mathcal{F}} |f(x) - f(y)| \leq \sup_{f \in \mathcal{F}} \varepsilon = \varepsilon.$$

This shows that $s(x)$ is continuous and the proof is complete. $\square$

Our only application of this lemma will be the following.

Proposition 2.2. *Let $U \subseteq \mathbb{R}^N$ be non-empty and let $f \in L^1(\mathbb{R}^N)$. For $r \geq 0$, we define the Lévy concentration function of f as*

$$Q(r) := \sup_{y \in U} \int_{B(y, r)} |f|. \quad (2.1)$$

The function $Q(r)$ is continuous on $[0, \infty)$.

Proof. For fixed $y \in U$, we define

$$Q_y : [0, \infty) \rightarrow \mathbb{R}, \quad r \mapsto \int_{B(y, r)} |f|.$$

Since $f \in L^1(\mathbb{R}^N)$, each Q_y is continuous at 0, i.e.

$$\lim_{r \searrow 0} Q_y(r) = 0.$$

By virtue of Lemma 2.1, it is enough to show that the family $\mathcal{F} := \{Q_y\}_{y \in U}$ is equicontinuous and equibounded on any compact interval $[a, b]$. To see that this family is equibounded, let $y \in U$ and $r \geq 0$ be arbitrary. Clearly, one has the following:

$$|Q_y(r)| = \int_{B(y, r)} |f| \leq \int_{\mathbb{R}^N} |f| < \infty.$$

It follows that $\mathcal{F}$ is equibounded on $[0, \infty)$, and hence on $[a, b]$. Next, we establish the equicontinuity of $\mathcal{F}$ on $[a, b]$. Let $\varepsilon > 0$ be given. Since $f \in L^1(\mathbb{R}^N)$, there exists $\delta > 0$ such that

$$\int_E |f| < \varepsilon$$

whenever $E \subset \mathbb{R}^N$ is Lebesgue measurable with $m(E) < \delta$.² Now, for any $r, q \in [a, b]$ and every $y \in U$, it is easily seen that

$$|Q_y(r) - Q_y(q)| = \left| \int_{B(y,r)} |f| - \int_{B(y,q)} |f| \right| = \int_{B(y,r) \setminus B(y,q)} |f|$$

where we are assuming, without loss of generality, that $r \geq q$. Letting ω_n be the volume of the unit ball in $\mathbb{R}^N$, we have

$$\begin{aligned} m(B(y,r) \setminus B(y,q)) &= \omega_n(r^n - q^n) \\ &= \omega_n(r^{n-1} + r^{n-2}q + \cdots + rq^{n-2} + q^{n-1})(r - q) \\ &\leq C(r - q), \end{aligned}$$

for a constant $C > 0$ independent of $r, q \in [a, b]$.³ Therefore, we will have

$$m(B(y,r) \setminus B(y,q)) < \delta$$

whenever $0 \leq (r - q) < \frac{\delta}{C}$. This implies that

$$|Q_y(r) - Q_y(q)| \leq \int_{B(y,r) \setminus B(y,q)} |f| < \varepsilon$$

for any such $r, q \in [a, b]$. This shows that the family $\mathcal{F}$ is equicontinuous and equibounded on every $[a, b] \subset [0, \infty)$ and this completes the proof. $\square$

2.1 Understanding Weak Convergence in $\mathcal{D}^{1,p}(\mathbb{R}^N)$

As mentioned previously, the dual space of $\mathcal{D}^{1,p}(\mathbb{R}^N)$ will play an important role in the proof of Theorem 1. Consequently, we seek to properly understand weak convergence in $\mathcal{D}^{1,p}(\mathbb{R}^N)$. Of course, since strong convergence in $\mathcal{D}^{1,p}(\mathbb{R}^N)$ only tests the L^p -convergence of the gradients, we should expect that weak convergence in $\mathcal{D}^{1,p}(\mathbb{R}^N)$ be characterized by the weak convergence of the gradients in L^p . As it turns out, this is indeed the case.

²Here, $m(\cdot)$ denotes the Lebesgue measure on $\mathbb{R}^N$.

³Note that this constant C will depend on the interval $[a, b]$.

Proposition 2.3. *Let (v_n) be a sequence in $\mathcal{D}^{1,p}(\mathbb{R}^N)$ and fix $v \in \mathcal{D}^{1,p}(\mathbb{R}^N)$. Then, $v_n \rightharpoonup v$ in $\mathcal{D}^{1,p}(\mathbb{R}^N)$ if and only if $\nabla v_n \rightharpoonup \nabla v$ in $L^p(\mathbb{R}^N; \mathbb{R}^N)$.*

Proof. We begin by defining the following map

$$T : \mathcal{D}^{1,p}(\mathbb{R}^N) \rightarrow L^p(\mathbb{R}^N; \mathbb{R}^N), \quad u \mapsto \nabla u.$$

Clearly, T is a linear isometry and is hence an embedding. Through T , we can identify $\mathcal{D}^{1,p}(\mathbb{R}^N)$ with the subspace $T(\mathbb{R}^N)$ of $L^p(\mathbb{R}^N; \mathbb{R}^N)$. As a first step, we will show that $v_n \rightharpoonup v$ in $\mathcal{D}^{1,p}(\mathbb{R}^N)$ if and only if $\nabla v_n \rightharpoonup \nabla v$ in $T(\mathbb{R}^N)$. Let us now assume that v_n converges weakly to v in $\mathcal{D}^{1,p}(\mathbb{R}^N)$. Given a continuous linear functional ϕ on $T(\mathbb{R}^N)$, we define

$$\varphi : \mathcal{D}^{1,p}(\mathbb{R}^N) \rightarrow \mathbb{R}, \quad \varphi(u) := \phi(\nabla u).$$

In other words, we put $\varphi := \phi \circ T$. Clearly, φ is a continuous linear functional on $\mathcal{D}^{1,p}(\mathbb{R}^N)$. Since v_n converges weakly to v in $\mathcal{D}^{1,p}(\mathbb{R}^N)$, we must have $\varphi(v_n) \rightarrow \varphi(v)$ as $n \rightarrow \infty$. However, this is equivalent to the statement

$$\lim_{n \rightarrow \infty} \phi(\nabla v_n) = \phi(\nabla v).$$

This shows that ∇v_n converges weakly to ∇v in $L^p(\mathbb{R}^N; \mathbb{R}^N)$, as $n \rightarrow \infty$. Conversely, we assume that $\nabla v_n \rightharpoonup \nabla v$ in $T(\mathbb{R}^N)$, and fix a continuous linear functional φ on $\mathcal{D}^{1,p}(\mathbb{R}^N)$. Next, consider the map

$$\phi : T(\mathbb{R}^N) \rightarrow \mathbb{R}, \quad \phi := \varphi \circ T^{-1}.$$

Again, it is obvious that ϕ is a continuous linear functional on $T(\mathbb{R}^N)$. Since ∇v_n converges weakly to ∇v in $T(\mathbb{R}^N)$, we see that $\phi(\nabla v_n)$ converges to $\phi(\nabla v)$. By definition, this is equivalent to having

$$\lim_{n \rightarrow \infty} \varphi(v_n) = \varphi(v).$$

To summarize, we have shown that weak convergence in $\mathcal{D}^{1,p}(\mathbb{R}^N)$ is equivalent to the weak convergence of the gradients in $T(\mathbb{R}^N)$. To complete the proof, we need only show that weak convergence of the gradients in $T(\mathbb{R}^N)$ is equivalent to their weak convergence in all of $L^p(\mathbb{R}^N)$. Here, the only non-trivial claim is that weak convergence in $L^p(\mathbb{R}^N; \mathbb{R}^N)$ implies weak convergence in $T(\mathbb{R}^N)$. However, this is a simple consequence of the Hahn-Banach theorem. Indeed, any continuous linear functional ϕ on $T(\mathbb{R}^N)$ can be extended to a continuous linear functional ψ on all of $L^p(\mathbb{R}^N; \mathbb{R}^N)$. Thus, if $\nabla v_n \rightharpoonup \nabla v$ in $L^p(\mathbb{R}^N; \mathbb{R}^N)$, then

$$\psi(v_n) \rightarrow \psi(v) \quad \text{in } L^p(\mathbb{R}^N; \mathbb{R}^N) \text{ as } n \rightarrow \infty.$$

Clearly, this simply means that $\lim_{n \rightarrow \infty} \phi(v_n) = \phi(v)$ whence we have established that ∇v_n converges weakly to ∇v in $T(\mathbb{R}^N)$. This completes the proof. $\square$

This technical result will only be used when establishing the following result.

Corollary 2.4. *Let (v_n) be a sequence that is uniformly bounded in both $L^{p^*}(\mathbb{R}^N)$ and $\mathcal{D}^{1,p}(\mathbb{R}^N)$. There exists $v \in \mathcal{D}^{1,p}(\mathbb{R}^N)$ and a subsequence of (v_n) that converges weakly to v in $\mathcal{D}^{1,p}(\mathbb{R}^N)$.*

Remark 2.1. Note that we cannot directly apply any familiar results about weak compactness in Banach spaces as $\mathcal{D}^{1,p}(\mathbb{R}^N)$ may not be complete or reflexive, a priori. However, as will become apparent in the proof, we will be “saved” by the familiar properties of L^p -spaces. Indeed, the requirement $p \in (1, \infty)$ implies that $p' \in (1, \infty)$. Therefore, duality tells us that

$$(L^p(\mathbb{R}^N; \mathbb{R}^N))^{**} \cong (L^{p'}(\mathbb{R}^N; \mathbb{R}^N))^* \cong L^p(\mathbb{R}^N; \mathbb{R}^N)$$

whence we see that $L^p(\mathbb{R}^N; \mathbb{R}^N)$ is reflexive. If $\mathcal{X}$ is a Banach space, then $\mathcal{X}$ is reflexive if and only if the closed unit ball in $\mathcal{X}$ is weakly compact⁴. It therefore follows that bounded sequences in $L^p(\mathbb{R}^N; \mathbb{R}^N)$ have a weakly convergent subsequence. Since $W^{1,p}(\Omega)$ is reflexive for $1 < p < \infty$, the reasoning shows that bounded sequences in $W^{1,p}(\Omega)$ have weakly convergent subsequences.

Proof of Corollary. By definition of the $\mathcal{D}^{1,p}(\mathbb{R}^N)$ -norm, the sequence of gradients (∇v_n) is bounded in $L^p(\mathbb{R}^N; \mathbb{R}^N)$. By basic L^p -theory, after passing to a subsequence, we may assume that ∇v_n converges weakly to some $w = (w_1, \dots, w_N)$ in $L^p(\mathbb{R}^N; \mathbb{R}^N)$. Passing to yet another subsequence, we may also assume that v_n converges weakly to $v \in L^{p^*}(\mathbb{R}^N)$. Next, we claim that $v \in \mathcal{D}^{1,p}(\mathbb{R}^N)$ and that $\nabla v = w$. To see this, fix an index $i = 1, \dots, N$ and let $\varphi \in C_c^\infty(\mathbb{R}^N)$ be given. Weak convergence in $L^{p^*}(\mathbb{R}^N)$ gives

$$\int_{\mathbb{R}^N} v \partial_i \varphi = \lim_{n \rightarrow \infty} \int_{\mathbb{R}^N} v_n \partial_i \varphi = - \lim_{n \rightarrow \infty} \int_{\mathbb{R}^N} \varphi \partial_i v_n.$$

Since ∇v_n converges weakly to w in $L^p(\mathbb{R}^N; \mathbb{R}^N)$, it is not hard to check that also $\partial_i v_n$ converges weakly to w_i in $L^p(\mathbb{R}^N)$. Using this with the above, we discover that

$$\int_{\mathbb{R}^N} v \partial_i \varphi = - \lim_{n \rightarrow \infty} \int_{\mathbb{R}^N} \varphi \partial_i v_n = - \int_{\mathbb{R}^N} \varphi w_i.$$

We conclude that $v \in \mathcal{D}^{1,p}(\mathbb{R}^N)$ and that $\nabla v = w$. Finally, since $\nabla v_n \rightarrow w = \nabla v$ weakly in $L^p(\mathbb{R}^N; \mathbb{R}^N)$, it follows from the previous lemma that v_n converges weakly to v in $\mathcal{D}^{1,p}(\mathbb{R}^N)$. $\square$

⁴The proof of this fact is essentially an application of the Banach-Alaoglu theorem.

Before starting the proof of Theorem 1, we would like to partially strengthen the conclusion of this corollary. Obviously, it would be ideal to extract a subsequence that converges strongly in $\mathcal{D}^{1,p}(\mathbb{R}^N)$. Instead, we will have to be content with the following pointwise result.

Corollary 2.5. *In the context of Corollary 2.4 and its proof, there is a further subsequence that converges pointwise almost everywhere in $\mathbb{R}^N$ to v .*

Proof. For each $k \in \mathbb{N}$ we denote by B_k the open ball $B(0, k)$ in $\mathbb{R}^N$. For a fixed k , we consider the restrictions of the sequence (v_n) to B_k . In doing so, we obtain a bounded sequence in $W^{1,p}(B_k)$, for each $k \in \mathbb{N}$. By the Rellich-Kondrachov theorem, the embedding

$$W^{1,p}(B_k) \hookrightarrow L^p(B_k)$$

is compact. Thus, passing to a subsequence if necessary, we may assume that

- (i) v_n converges weakly to some $u_k \in W^{1,p}(B_k)$ as $n \rightarrow \infty$;
- (ii) v_n converges *strongly* to u_k in $L^p(B_k)$ as $n \rightarrow \infty$; and
- (iii) v_n converges pointwise almost everywhere to u_k on B_k as $n \rightarrow \infty$.

By duality and the fact that $v_n \rightharpoonup v$ in $\mathcal{D}^{1,p}(\mathbb{R}^N)$, we have by Proposition 2.3 that

$$\lim_{n \rightarrow \infty} \int_{\mathbb{R}^N} g \cdot \nabla v_n = \int_{\mathbb{R}^N} g \cdot \nabla v,$$

for all $g \in L^{p'}(\mathbb{R}^N; \mathbb{R}^N)$. In particular,

$$\lim_{n \rightarrow \infty} \int_{B_k} g \cdot \nabla v_n = \int_{B_k} g \cdot \nabla v$$

for all $g \in L^p(B_k; \mathbb{R}^N)$. On the other hand, (i) ensures that

$$\lim_{n \rightarrow \infty} \int_{B_k} \nabla v_n \cdot g = \lim_{n \rightarrow \infty} \int_{B_k} \nabla u_k \cdot g$$

for all $g \in L^{p'}(B_k)$. Combining these last two equations shows that

$$\int_{B_k} (\nabla u_k - \nabla v) \cdot g = \lim_{n \rightarrow \infty} \int_{B_k} (\nabla u_k - \nabla v_n) \cdot g = 0$$

for all such g . By density, it follows that $u_k = v + \alpha_k$ on B_k for some $\alpha_k \in \mathbb{R}$. In other words, v_k converges pointwise almost everywhere to $v + \alpha_k$ on B_k , as $n \rightarrow \infty$. To complete the proof, we will now make use of a diagonal argument. For this family of balls $\{B_k\}_{k \geq 1}$, we inductively construct subsequences $\{v_{n,k}\}_{k \geq 1}$ for which the above holds true. Therefore, there is a corresponding sequence $(\alpha_k)_{k \geq 1}$ of real numbers such that

$$v_{n,k} \rightarrow v + \alpha_k \quad \text{pointwise a.e. on } B_k \text{ as } n \rightarrow \infty,$$

for each k . Finally, for each $k \in \mathbb{N}$, an application of Fatou's lemma yields

$$\begin{aligned} \alpha_k m(B_k)^{p^*} &\leq \liminf_{n \rightarrow \infty} \int_{B_k} |v_{n,k} - v|^{p^*} \leq \limsup_{n \rightarrow \infty} \int_{B_k} (|v_{n,k}| + |v|)^{p^*} \\ &\leq 2^{p^*-1} \limsup_{n \rightarrow \infty} \left(\int_{B_k} |v_{n,k}|^{p^*} + \int_{B_k} |v|^{p^*} \right) \\ &\leq 2^{p^*-1} \sup_{n \in \mathbb{N}} \left(\int_{\mathbb{R}^N} |v_{n,k}|^{p^*} + \int_{\mathbb{R}^N} |v|^{p^*} \right) \end{aligned}$$

where this last term is uniformly bounded in n, k since (v_n) is bounded in $W^{1,p}(\mathbb{R}^N)$ and $v \in L^{p^*}(\mathbb{R}^N)$. Since $m(B_k) \rightarrow \infty$ as $k \rightarrow \infty$, we conclude that $\alpha_k \rightarrow 0$ as $k \rightarrow \infty$. Then, the diagonal subsequence $(v_{n,n})_{n \in \mathbb{N}}$ will converge pointwise almost everywhere to v on $\mathbb{R}^N$ as $n \rightarrow \infty$, which proves the claim. $\square$

3 The Proof of Theorem 1

We are now ready to illustrate the method used by Mercuri-Willem in the proof of Theorem 1. In the spirit of their proof, we shall divide the proof into several steps for the sake of readability. Recall that $\|\cdot\|$ by default denotes the $\mathcal{D}^{1,p}(\mathbb{R}^N)$ norm and **not** the $W_0^{1,p}(\Omega)$ -norm.

Step 1. *There is a subsequence, also denoted (u_n) , and $v_0 \in W_0^{1,p}(\Omega)$ such that $u_n \rightharpoonup v_0$ in $W_0^{1,p}(\Omega)$ and strongly in $L^p(\Omega)$, u_n converges pointwise almost everywhere to v_0 on Ω , and $D\Phi(v_0) = 0$. Furthermore, the sequence in $W_0^{1,p}(\Omega)$ given by*

$$u_n^1 := u_n - v_0$$

is uniformly bounded in $W_0^{1,p}(\Omega)$ and satisfies

- (1) $\|u_n^1\|^p = \|u_n\|^p - \|v_0\|^p + o(1)$;
- (2) $\lim_{n \rightarrow \infty} \Phi_\infty(u_n^1) = \lim_{n \rightarrow \infty} \Phi(u_n) - \Phi(v_0)$; and

(3) $D\Phi_\infty(u_n^1) \rightarrow 0$ in $W^{-1,p'}(\Omega)$ as $n \rightarrow \infty$.

Proof of Step 1. Since the sequence (u_n) is bounded in $W_0^{1,p}(\Omega)$, we may obviously extract a subsequence converging weakly to some v_0 in $W_0^{1,p}(\Omega)$. Furthermore, since Ω is a smoothly bounded domain, the embedding

$$W_0^{1,p}(\Omega) \hookrightarrow L^p(\Omega)$$

is compact by the Rellich-Kondrachov theorem. Passing to yet another subsequence, we might as well assume that u_n converges *strongly* to some function w in $L^p(\Omega)$ and that $\lim_{n \rightarrow \infty} u_n(x) = w(x)$ for almost every $x \in \Omega$. In particular, u_n converges weakly to w in $L^p(\Omega)$. However, by identifying $W_0^{1,p}(\Omega)$ with a subspace of $L^p(\Omega)$, it is clear that weak convergence in $W_0^{1,p}(\Omega)$ implies weak convergence in $L^p(\Omega)$. This means that $u_n \rightharpoonup v_0$ in $L^p(\Omega)$; by the uniqueness of weak limits we must then have $v_0 = w$ almost everywhere in Ω .

By Theorem B, it is immediate that $D\Phi(v_0) = 0$. Consequently, it remains only to verify the asymptotic identities in (1)-(3). This turns out to be quite easy since Theorem B directly implies both (2) and (3). As for (1), we have by this same theorem that

$$\lim_{n \rightarrow \infty} (\|u_n\|^p - \|u_n^1\|^p) = \|v_0\|^p.$$

It follows that

$$\begin{aligned} \|u_n^1\|^p + \|v_0\|^p - \|u_n\|^p &= \|v_0\|^p - (\|u_n\|^p - \|u_n^1\|^p) \\ &= \|v_0\|^p - \|v_0\|^p + o(1) = o(1). \end{aligned}$$

This completes the proof. We also point out that (u_n^1) is a sequence in $W_0^{1,p}(\Omega)$ which may be considered as a sequence in $W^{1,p}(\mathbb{R}^N)$. In fact, since the sequence $(u_n)_{n \in \mathbb{N}}$ is a Palais-Smale sequence, the auxiliary sequence $(u_n^1)_{n \in \mathbb{N}}$ will be uniformly bounded in $W_0^{1,p}(\mathbb{R}^N)$. $\square$

Step 2. *Theorem 1 holds true in the case where $u_n^1 \rightarrow 0$ strongly in $L^{p^*}(\Omega)$.*

Proof of Step 2. By virtue of Step 1, we know that $D\Phi_\infty(u_n^1)$ converges to 0 in $W^{-1,p'}(\Omega)$ as $n \rightarrow \infty$. In particular, we know that

$$\Phi'_\infty(u_n^1)(h) = \int_{\mathbb{R}^N} \left(|\nabla u_n^1|^{p-2} \nabla u_n^1 \cdot \nabla h - \mu(u_n^1)_{+}^{p^*-1} h \right) \rightarrow 0, \quad \text{as } n \rightarrow \infty,$$

for every fixed $h \in W_0^{1,p}(\Omega)$. Note that this expression makes sense as $W_0^{1,p}(\Omega)$ can be interpreted as a subspace of $\mathcal{D}^{1,p}(\mathbb{R}^N)$ by the observation following Remark 1.1. Furthermore, for every integer $n \geq 1$ there holds

$$\begin{aligned} |\Phi'_\infty(u_n^1)(u_n^1)| &\leq \sup_{n \in \mathbb{N}} \|u_n^1\| \|\Phi'_\infty(u_n^1)\|_{W^{-1,p'}(\Omega)} \\ &\leq \sup_{n \in \mathbb{N}} \|u_n^1\|_{W^{1,p}(\Omega)} \|\Phi'_\infty(u_n^1)\|_{W^{-1,p'}(\Omega)} \rightarrow 0, \end{aligned}$$

as $n \rightarrow \infty$. Here, we have used the uniform boundedness of the (u_n^1) in $W_0^{1,p}(\Omega)$. Returning to our expression for $\Phi'_\infty(u_n^1)(h)$, we see that

$$\Phi'_\infty(u_n^1)(u_n^1) = \int_{\mathbb{R}^N} \left(|\nabla u_n^1|^p - \mu(u_n^1)_+^{p^*-1} u_n^1 \right) \rightarrow 0, \quad \text{as } n \rightarrow \infty.$$

However, $u_n^1 \rightarrow 0$ in $L^{p^*}(\Omega)$ would imply that, as $n \rightarrow \infty$,

$$\begin{aligned} \left| \int_{\mathbb{R}^N} \mu(u_n^1)_+^{p^*-1} u_n^1 \right| &\leq \int_{\mathbb{R}^N} \mu |u_n^1| (u_n^1)_+^{p^*-1} \\ &\leq \int_{\mathbb{R}^N} \mu |u_n^1|^{p^*} \rightarrow 0. \end{aligned}$$

We then infer that

$$\int_{\mathbb{R}^N} |\nabla u_n^1|^p \rightarrow 0, \quad \text{as } n \rightarrow \infty.$$

In other words, $\nabla u_n \rightarrow \nabla v_0$ in $L^p(\Omega)$. From this we see that

$$\|u_n - v_0\| \rightarrow 0 \quad \text{and} \quad \|u_n\| \rightarrow \|v_0\|$$

as $n \rightarrow \infty$. Since $u_n \rightarrow v_0$ in $L^p(\Omega)$ by Step 1, we see that $u_n^1 \rightarrow 0$ in $L^p(\mathbb{R}^N)$. Combining these facts shows that $u_n^1 \rightarrow 0$ in $W_0^{1,p}(\Omega)$. That is, we have $\lim u_n = v_0$ in $W_0^{1,p}(\Omega)$. Furthermore, Step 1 also states that $\Phi'(v_0) = 0$. Therefore,

$$\Phi'(v_0)(h) = 0, \quad \forall h \in W_0^{1,p}(\Omega).$$

In particular, this holds for all $h \in C_c^\infty(\Omega)$. It follows that v_0 solves problem (1.3). With $k = 0$, we will be done if $v_0 \geq 0$ in Ω . Indeed, after passing to another subsequence, we may assume that $(u_n)_-$ converges pointwise almost everywhere to 0 on Ω . Since u_n converges to v_0 pointwise a.e. on Ω , the claim follows. $\square$

Next, we will consider a family of Lévy concentration functions. For every index $n \geq 1$, we define a real valued function Q_n on $[0, \infty)$ via the formula

$$Q_n(r) := \sup_{y \in \Omega} \int_{B(y,r)} |u_n^1|^{p^*}.$$

Note that each Q_n is well defined since every $u_n^1 \in L^{p^*}(\mathbb{R}^N)$ by the Sobolev embedding theorem. By the same logic, since the $\{u_n^1\}_{n \in \mathbb{N}}$ are uniformly bounded in $W_0^{1,p}(\mathbb{R}^N)$, the family

$$\mathcal{F}_n := \{Q_n(\cdot)\}_{n \in \mathbb{N}}$$

defines (by Proposition 2.2) a collection of continuous functions $\{f_n\}_{n \in \mathbb{N}}$ on $[0, \infty)$. Now, in light of Step 2, we may assume that $u_n^1 \not\rightarrow 0$ strongly in $L^{p^*}(\Omega)$. After passing to a subsequence, this means that there exists $\delta > 0$ such that

$$\inf_{n \in \mathbb{N}} \int_{\Omega} |u_n^1|^{p^*} > \delta.$$

Without loss of generality, we may assume that

$$0 < \delta < \left(\frac{S_p}{2\mu} \right)^{N/p}$$

where $S_p > 0$ is the best constant for which the Sobolev inequality

$$S_p \|w\|_{L^{p^*}(\mathbb{R}^N)} \leq \|w\|_{W^{1,p}(\mathbb{R}^N)}, \quad \forall w \in W^{1,p}(\mathbb{R}^N),$$

holds true. Now, every Q_n is a continuous function on $[0, \infty)$ satisfying

$$Q_n(0) = 0 \quad \text{and} \quad \lim_{r \rightarrow \infty} Q_n(r) > \delta.$$

By the intermediate value theorem, there exists $r_n \in [0, \infty)$ such that $Q_n(r_n) = \delta$. Letting R_n be the set of all such r_n , we put $\lambda_n^1 := \inf R_n \geq 0$. Let now $(r_l)_{l \in \mathbb{N}}$ be a sequence in R_n converging to λ_n^1 as $l \rightarrow \infty$. By continuity, it is clear that

$$Q_n(\lambda_n^1) = \lim_{l \rightarrow \infty} Q_n(r_l) = \delta.$$

Furthermore, since $Q_n(0) = 0$, we have $\lambda_n^1 > 0$. Proceeding inductively, we obtain a sequence (λ_n^1) in $(0, \infty)$ such that

$$Q_n(\lambda_n^1) = \sup_{y \in \Omega} \int_{B(y, \lambda_n^1)} |u_n^1|^{p^*} = \delta,$$

for each $n \in \mathbb{N}$. Fixing $n \in \mathbb{N}$, we may find a sequence $(\xi_{m,n}^1)_{m \in \mathbb{N}}$ in $\bar{\Omega}$ such that

$$\delta = \lim_{m \rightarrow \infty} \int_{B(\xi_{m,n}^1, \lambda_n)} |u_n^1|^{p^*}.$$

Passing to a subsequence if necessary, we may assume that $\xi_{m,n}^1 \rightarrow y_n^1 \in \bar{\Omega}$ as we let $m \rightarrow \infty$. In fact, we may assume that

$$\delta = \lim_{m \rightarrow \infty} \int_{B(\xi_{m,n}^1, \lambda_n)} |u_n^1|^{p^*} = \int_{B(y_n^1, \lambda_n^1)} |u_n^1|^{p^*}$$

for every index $n \in \mathbb{N}$. With this in mind, the following claim readily follows:

Step 3. *For each $n \in \mathbb{N}$ consider the domain*

$$\Omega_n := \frac{1}{\lambda_n^1} (\Omega - y_n^1).$$

A simple calculation shows that $x \in \Omega_n$ if and only if

$$\lambda_n^1 x + y_n^1 \in \Omega.$$

Next, we define a sequence $(v_n^1)_{n \in \mathbb{N}}$ in $W_0^{1,p}(\Omega_n) \subseteq W^{1,p}(\mathbb{R}^N)$ by the rule

$$v_n^1(x) := (\lambda_n^1)^{\frac{N-p}{p}} u_n^1(\lambda_n^1 x + y_n^1).$$

The family $\{v_n^1\}_{n \in \mathbb{N}}$ belongs to the space $W^{1,p}(\mathbb{R}^N)$ and is uniformly bounded in $\mathcal{D}^{1,p}(\mathbb{R}^N)$. Moreover, one has

$$\delta = \sup_{y \in \bar{\Omega}_n} \int_{B(y,1)} |v_n^1|^{p^*} = \int_{B(0,1)} |v_n^1|^{p^*}. \quad (3.1)$$

Proof of Step 3. Since every v_n^1 is a rescaling of a function in $W^{1,p}(\mathbb{R}^N)$, it is clear that $\{v_n^1\}_{n \in \mathbb{N}} \subset W^{1,p}(\mathbb{R}^N) \subseteq \mathcal{D}^{1,p}(\mathbb{R}^N)$. To see that the $\{v_n^1\}_{n \in \mathbb{N}}$ is uniformly bounded in $\mathcal{D}^{1,p}(\mathbb{R}^N)$, it suffices to make the following change of variables:

$$\begin{aligned} \|v_n^1\|^p &= \|\nabla v_n^1\|_{L^p(\mathbb{R}^N)}^p = \int_{\mathbb{R}^N} \left| \lambda_n^1 \cdot (\lambda_n^1)^{\frac{N-p}{p}} \nabla u_n^1(\lambda_n^1 x + y_n^1) \right|^p dx \\ &= \int_{\mathbb{R}^N} \left| (\lambda_n^1)^{\frac{N}{p}} \nabla u_n^1(\lambda_n^1 x + y_n^1) \right|^p dx \\ &= \int_{\mathbb{R}^N} (\lambda_n^1)^N |\nabla u_n^1(\lambda_n^1 x + y_n^1)|^p dx \\ &= \int_{\mathbb{R}^N} |u_n^1|^{p^*} \\ &\leq \|u_n^1\|_{W^{1,p}(\mathbb{R}^N)}^p. \end{aligned}$$

Since the $\{u_n^1\}_{n \in \mathbb{N}}$ are uniformly bounded in $W^{1,p}(\mathbb{R}^N)$ by Step 1, the family $\{v_n^1\}_{n \in \mathbb{N}}$ is therefore bounded in $\mathcal{D}^{1,p}(\mathbb{R}^N)$. Next, we verify (3.1). Given $n \in \mathbb{N}$, we have by construction that

$$\begin{aligned} \int_{B(0,1)} |v_n^1|^{p^*} &= \int_{B(0,1)} \left| (\lambda_n^1)^{\frac{N-p}{p}} u_n^1(\lambda_n^1 x + y_n^1) \right|^{\frac{Np}{N-p}} dx \\ &= \int_{B(0,1)} (\lambda_n^1)^N |u_n^1(\lambda_n^1 x + y_n^1)|^{p^*} dx \\ &= \int_{B(y_n^1, \lambda_n^1)} |u_n^1|^{p^*} = \delta. \end{aligned}$$

Since $y \in \overline{\Omega_n}$ if and only if

$$y = \frac{z - y_n^1}{\lambda_n^1}$$

for a unique $z \in \bar{\Omega}$, a similar argument yields

$$\begin{aligned} \sup_{y \in \overline{\Omega_n}} \int_{B(y,1)} |v_n^1|^{p^*} &= \sup_{y \in \overline{\Omega_n}} \int_{B(y,1)} \left| (\lambda_n^1)^{\frac{N-p}{p}} u_n^1(\lambda_n^1 x + y_n^1) \right|^{\frac{Np}{N-p}} dx \\ &= \sup_{y \in \overline{\Omega_n}} \int_{B(y,1)} (\lambda_n^1)^N |u_n^1(\lambda_n^1 x + y_n^1)|^{p^*} dx \\ &= \sup_{z \in \bar{\Omega}} \int_{B(z, \lambda_n^1)} |u_n^1|^{p^*} = \delta. \end{aligned}$$

This directly implies (3.1) and the proof is complete. $\square$

Remark 3.1. Since Ω is bounded, the sequence of radii (λ_n^1) must also be bounded. To be more explicit, we must have $0 < \lambda_n^1 \leq 2 \operatorname{diam}(\Omega)$ for each $n \geq 1$. Furthermore, the sequence of centers (y_n^1) lives in $\bar{\Omega}$ and must also be bounded. Thus, after passing to another subsequence, we may find y^1 in $\bar{\Omega}$ and $\lambda^1 \geq 0$ such that

$$\lim_{n \rightarrow \infty} y_n^1 = y^1 \quad \text{and} \quad \lim_{n \rightarrow \infty} \lambda_n^1 = \lambda^1.$$

For each $n \in \mathbb{N}$, let us now consider the linear functional $D\Phi_\infty(u_n^1)(\cdot)$ on $\mathcal{D}^{1,p}(\mathbb{R}^N)$. Since $W_0^{1,p}(\Omega)$ is a closed subspace of $\mathcal{D}^{1,p}(\mathbb{R}^N)$, every $D\Phi_\infty(u_n^1)(\cdot)$ restricts to an element of the dual $W^{-1,p'}(\Omega)$. By a duality argument, we can find $f_n^1, \dots, f_n^N \in L^{p'}(\Omega)$

such that⁵

$$D\Phi_\infty(u_n^1)(h) = \sum_{i=1}^N \int_{\Omega} f_n^i \partial_i h$$

for all $h \in W_0^{1,p}(\Omega)$. Define

$$g_n^i = (\lambda_n^1)^{N-\frac{N}{p}} f_n^i (\lambda_n^1 x + y_n^1)$$

for $i = 1, \dots, N$.

Step 4. *Adhering to the notation and terminology developed above, one has*

$$D\Phi_\infty(v_n^1)(h) = \sum_{i=1}^N \int_{\Omega_n} g_n^i \partial_i h$$

for all $h \in W_0^{1,p}(\Omega_n)$.

Proof of Step 4. First, we fix a function $h \in W_0^{1,p}(\Omega_n)$ and define

$$\tilde{h}(z) = h\left(\frac{z - y_n^1}{\lambda_n^1}\right).$$

Notice that $\tilde{h} \in W_0^{1,p}(\Omega)$. Furthermore, we have the following equalities;

$$\lambda_n^1 \partial_i \tilde{h}(z) = \partial_i h\left(\frac{z - y_n^1}{\lambda_n^1}\right) \quad \text{and} \quad \lambda_n^1 \nabla \tilde{h}(z) = \nabla h\left(\frac{z - y_n^1}{\lambda_n^1}\right).$$

Similarly, we see that

$$\nabla v_n^1(x) = (\lambda_n^1)^{(N-p)/p+1} \nabla u_n^1(\lambda_n^1 x + y_n^1) = (\lambda_n^1)^{N/p} \nabla u_n^1(\lambda_n^1 x + y_n^1).$$

⁵Given general domain Ω and a continuous linear functional φ on $W_0^{1,p}(\Omega)$, we can find $f, 0, f_1, \dots, f_N$ belonging to $L^{p'}(\Omega)$ such that

$$\varphi(w) = \int_{\Omega} f_0 w + \sum_{i=1}^N \int_{\Omega} f_i \partial_i w$$

for all $w \in W_0^{1,p}(\Omega)$. However, when Ω is bounded, one can dispense with the f_0 -term in the above. That is, we have $\varphi(w) = \sum_{i=1}^N \int_{\Omega} f_i^p \partial_i w$ for suitable $f_1, \dots, f_N$ in $L^{p'}(\Omega)$. See Adams-Fournier [1] and Brézis [2] for a precise statement and proof.

Fix now an integer $n \geq 1$ and let $h \in W_0^{1,p}(\Omega_n)$ be given. It is easily seen that

$$\begin{aligned}
\sum_{i=1}^N \int_{\Omega_n} g_n^i \partial_i h &= \sum_{i=1}^N \int_{\Omega_n} (\lambda_n^1)^{N-\frac{N}{p}} f_n^i(\lambda_n^1 x + y_n^1) \partial_i h(x) \, dx \\
&= (\lambda_n^1)^{N-\frac{N}{p}-N} \sum_{i=1}^N \int_{\Omega} f_n^i(z) \partial_i h \left(\frac{z - y_n^1}{\lambda_n^1} \right) \, dz \\
&= (\lambda_n^1)^{-\frac{N}{p}} \sum_{i=1}^N \int_{\Omega} f_n^i(z) \partial_i h \left(\frac{z - y_n^1}{\lambda_n^1} \right) \, dz \\
&= (\lambda_n^1)^{1-N/p} \sum_{i=1}^N \int_{\Omega} f_n^i(z) \partial_i \tilde{h}(z) \, dz \\
&= (\lambda_n^1)^{1-N/p} D\Phi_{\infty}(u_n^1)(\tilde{h}).
\end{aligned}$$

On the other hand, $D\Phi_{\infty}(v_n^1)(h)$ is given by

$$\int_{\mathbb{R}^N} \left[|\nabla v_n^1|^{p-2} \nabla v_n^1 \cdot \nabla h - \mu (v_n^1)_+^{p^*-1} h \right]$$

which, when expanded, becomes

$$\begin{aligned}
&\int_{\mathbb{R}^N} |(\lambda_n^1)^{N/p} \nabla u_n^1(\lambda_n^1 x + y_n^1)|^{p-2} (\lambda_n^1)^{N/p} \nabla u_n^1(\lambda_n^1 x + y_n^1) \cdot \nabla h(x) \, dx - \int_{\mathbb{R}^N} \mu (v_n^1)_+^{p^*-1} h \\
&= (\lambda_n^1)^{N(p-1)/p} \int_{\mathbb{R}^N} |\nabla u_n^1(\lambda_n^1 x + y_n^1)|^{p-2} \nabla u_n^1(\lambda_n^1 x + y_n^1) \cdot \nabla h(x) \, dx \\
&\quad - \int_{\mathbb{R}^N} \mu ((\lambda_n^1)^{(N-p)/p} u_n^1(\lambda_n^1 x + y_n^1))_+^{p^*-1} h(x) \, dx \\
&= (\lambda_n^1)^{N(p-1)/p-N} \int_{\mathbb{R}^N} |\nabla u_n^1(z)|^{p-2} \nabla u_n^1(z) \cdot \nabla h \left(\frac{z - y_n^1}{\lambda_n^1} \right) \, dz \\
&\quad - (\lambda_n^1)^{(N-p)(p^*-1)/p} \int_{\mathbb{R}^N} \mu (u_n^1(\lambda_n^1 x + y_n^1))_+^{p^*-1} h(x) \, dx \\
&= (\lambda_n^1)^{-N/p} \int_{\mathbb{R}^N} |\nabla u_n^1(z)|^{p-2} \nabla u_n^1(z) \cdot \nabla h \left(\frac{z - y_n^1}{\lambda_n^1} \right) \, dz \\
&\quad - (\lambda_n^1)^{(N-p)(p^*-1)/p-N} \int_{\mathbb{R}^N} \mu (u_n^1(z))_+^{p^*-1} h \left(\frac{z - y_n^1}{\lambda_n^1} \right) \, dz \\
&= (\lambda_n^1)^{1-N/p} \int_{\mathbb{R}^N} |\nabla u_n^1(z)|^{p-2} \nabla u_n^1(z) \cdot \nabla \tilde{h}(z) \, dz - (\lambda_n^1)^{1-N/p} \int_{\mathbb{R}^N} \mu (u_n^1(z))_+^{p^*-1} \tilde{h}(z) \, dz.
\end{aligned}$$

After some cleaning up, this last expression is equal to

$$(\lambda_n^1)^{1-N/p} \int_{\mathbb{R}^N} \left[|\nabla u_n^1(z)|^{p-2} \nabla u_n^1 \cdot \nabla \tilde{h} - \mu (u_n^1)_+^{p^*-1} \tilde{h} \right],$$

which is precisely

$$(\lambda_n^1)^{1-N/p} D\Phi_\infty(u_n^1)(\tilde{h}).$$

Combining these identities, we find that for all $h \in W_0^{1,p}(\Omega_n)$:

$$D\Phi_\infty(v_n^1)(h) = (\lambda_n^1)^{1-N/p} D\Phi_\infty(u_n^1)(\tilde{h}) = \sum_{i=1}^N \int_{\Omega_n} g_n^i \partial_i h.$$

□

In light of the observations made in Remark 3.1, we may assume that $\Omega_n \rightarrow \Omega_\infty$, as $n \rightarrow \infty$, where $\Omega_\infty \subseteq \mathbb{R}^N$ is

- a rescaled translation of Ω if $\lambda^1 > 0$;
- all of $\mathbb{R}^N$ if $\lambda^1 = 0$ and $y^1 \in \Omega$; or
- a half space if $\lambda^1 = 0$ and $y^1 \in \partial\Omega$.

Here, the convergence should be understood in the sense of indicator functions. However, a geometric picture of this convergence is enough for our purposes.

Step 5. *The sequence (v_n^1) is uniformly bounded in $\mathcal{D}^{1,p}(\mathbb{R}^N)$ and, after passing to a subsequence, we may assume that $v_n^1 \rightharpoonup v_1$ for some $v_1 \in \mathcal{D}^{1,p}(\mathbb{R}^N)$ and that*

$$\lim_{n \rightarrow \infty} v_n^1(x) = v_1(x)$$

for almost every $x \in \mathbb{R}^N$.

Proof of Step 5. A simple change of variables shows that

$$\begin{aligned} \|v_n^1\|^p &= \int_{\mathbb{R}^N} |\nabla v_n^1|^p = \int_{\mathbb{R}^N} |(\lambda_n^1)^{N/p} \nabla u_n^1(\lambda_n^1 x + y_n^1)|^p dx \\ &= (\lambda_n^1)^N \int_{\mathbb{R}^N} |\nabla u_n^1(\lambda_n^1 x + y_n^1)|^p dx \\ &= \int_{\mathbb{R}^N} |\nabla u_n^1|^p \\ &\leq \|u_n^1\|_{W^{1,p}(\mathbb{R}^N)}^p. \end{aligned}$$

Since the sequence (u_n^1) is uniformly bounded in $W^{1,p}(\mathbb{R}^N)$, we infer that (v_n^1) is uniformly bounded in $\mathcal{D}^{1,p}(\mathbb{R}^N)$. The remaining assertions follow at once from Corollary 2.4 and Corollary 2.5. $\square$

Step 6. *The weak $\mathcal{D}^{1,p}(\mathbb{R}^N)$ limit v_1 of v_n^1 is non-zero.*

Proof of Step 6. We argue by contradiction. If $v_1 = 0$ in $\mathcal{D}^{1,p}(\mathbb{R}^N)$, then v_1 vanishes almost everywhere in $\mathbb{R}^N$. Recall that the $\{v_n^1\}_{n \in \mathbb{N}}$ are uniformly bounded in $\mathcal{D}^{1,p}(\mathbb{R}^N)$. By direct computation, we also find that

$$\begin{aligned} \int_{\mathbb{R}^N} |v_n^1|^{p^*} &= \int_{\mathbb{R}^N} (\lambda_n^1)^{\frac{N-p}{p} \cdot \frac{Np}{N-p}} |u_n^1(\lambda_n^1 x + y_n^1)|^{p^*} dx \\ &= \int_{\mathbb{R}^N} (\lambda_n^1)^N |u_n^1(\lambda_n^1 x + y_n^1)|^{p^*} dx \\ &= \int_{\mathbb{R}^N} |u_n^1|^{p^*}; \end{aligned}$$

where this last term can be bounded independently of n by the Sobolev embedding theorem. This shows that the $\{v_n^1\}_{n \in \mathbb{N}}$ also form a uniformly bounded sequence in $L^{p^*}(\mathbb{R}^N)$. Hence, given an open ball $B \subset \mathbb{R}^N$, we see that $\{v_n^1\}_{n \in \mathbb{N}}$ is a uniformly bounded sequence in $W^{1,p}(B)$. By the Rellich-Kondrachov theorem, we may assume that $v_n^1 \rightarrow w$ strongly in $L^p(B)$. Then, passing to a subsequence yet again, we must have $v_n^1 \rightarrow w$ pointwise almost everywhere in Ω . Hence, $w = v_1$ almost everywhere on Ω . In short, we get a subsequence v_n^1 that converges to 0 strongly in $L^p(B)$. By taking an exhaustion of $\mathbb{R}^N$ by open balls of radius $m \geq 1$ and using a diagonal argument, we find a subsequence (also denoted by (v_n^1)) that converges to 0 in $L^p(B_m)$, for each $m \in \mathbb{N}$. Clearly, this means that $v_n^1 \rightarrow 0$ in $L_{\text{loc}}^p(\mathbb{R}^N)$ as $n \rightarrow \infty$.

By using Step 1 together with a duality result from Brézis [2] (Proposition 9.20 therein), it follows that

$$\sum_{i=1}^N \int_{\Omega} |f_n^i|^{p'} = \sum_{i=1}^N \int_{\Omega_n} |g_n^i|^{p'} \rightarrow 0, \quad \text{as } n \rightarrow \infty. \quad (3.2)$$

Next, recall that Ω_{∞} denotes the “limiting domain” of the Ω_n . By definition, if $y \in \Omega_{\infty}$, then y belongs to Ω_n for all sufficiently large n . Fix now a point $y \in \Omega_{\infty}$ and consider the ball $B(y, 1)$, which may not be contained in Ω_{∞} . For a given test function $h \in C_c^{\infty}(\mathbb{R}^N)$

with $\text{supp}(h) \subset B(y, 1)$, Hölder's inequality with exponent $\frac{N}{p}$, we have

$$\begin{aligned} \int_{\mathbb{R}^N} |h|^p |v_n^1|^{p^*} &= \int_{\mathbb{R}^N} |v_n^1|^{p^*-p} \cdot |hv_n^1|^p \\ &\leq \left(\int_{\text{supp}(h)} |v_n^1|^{p^*} \right)^{p/N} \left(\int_{\mathbb{R}^N} |hv_n^1|^{p^*} \right)^{\frac{N-p}{N}}. \end{aligned}$$

From this, an application of Sobolev's inequality shows that

$$\int_{\mathbb{R}^N} |h|^p |v_n^1|^{p^*} \leq \left(\int_{\text{supp}(h)} |v_n^1|^{p^*} \right)^{p/N} \left(\int_{\mathbb{R}^N} |hv_n^1|^{p^*} \right)^{\frac{N-p}{N}} \quad (3.3)$$

$$\leq S_p^{-1} \left(\int_{\text{supp}(h)} |v_n^1|^{p^*} \right)^{p/N} \left(\int_{\mathbb{R}^N} |hv_n^1|^p \right). \quad (3.4)$$

Because v_n^1 converges *strongly* to 0 in $L_{\text{loc}}^p(\mathbb{R}^N)$, one has

$$\begin{aligned} \int_{\Omega_n} |h \nabla v_n^1|^p &= \int_{\mathbb{R}^N} |h|^p |\nabla v_n^1|^p + o(1) \\ &= \int_{\mathbb{R}^N} |\nabla v_n^1|^{p-2} \nabla v_n^1 \cdot \nabla (|h|^p v_n^1) + o(1) \\ &\leq \mu \int_{\mathbb{R}^N} |h|^p |v_n^1|^{p^*} + \sum_{i=1}^N \int_{\Omega_n} g_n^i \partial_i (|h|^p v_n^1) + o(1) \\ &\leq \mu \int_{B(y,1)} |h|^p |v_n^1|^{p^*} + o(1). \end{aligned}$$

From (3.2) and (3.3)-(3.4), it follows that

$$\begin{aligned} \int_{\Omega_n} |h \nabla v_n^1|^p &\leq \mu S_p^{-1} \delta^{\frac{p}{N}} \int_{\mathbb{R}^N} |\nabla (hv_n^1)|^p + o(1) \\ &\leq \frac{1}{2} \int_{\mathbb{R}^N} |\nabla (hv_n^1)|^p + o(1) \end{aligned}$$

whence we conclude that

$$\lim_{n \rightarrow \infty} \int_{\Omega_n} |h \nabla v_n^1|^p = 0.$$

It follows that $\nabla v_n^1 \rightarrow 0$ in $L_{\text{loc}}^p(B)$, for any ball B of radius 1 with center in Ω_∞ . After covering Ω_∞ with countably many balls, each of radius 1 with center in Ω_∞ , we infer

that ∇v_n^1 converges to 0 in $L_{\text{loc}}^p(\Omega_\infty)$, as $n \rightarrow \infty$. Similarly, $\|\nabla v_n^1\|_{L^p(B)} \rightarrow 0$ on any ball B compactly contained in the complement of Ω_∞ , as $n \rightarrow \infty$. To summarize, we have shown that $\nabla v_n^1 \rightarrow 0$ in $L_{\text{loc}}^p(\mathbb{R}^N)$ whence $v_n^1 \rightarrow 0$ in $W_{\text{loc}}^{1,p}(\mathbb{R}^N)$. Finally, the Sobolev inequality implies that

$$\lim_{n \rightarrow \infty} v_n^1 = 0 \quad \text{in } L_{\text{loc}}^{p^*}(\mathbb{R}^N),$$

which directly contradicts (3.1). $\square$

Step 7. $\lambda^1 = \lim_{n \rightarrow \infty} \lambda_n^1 = 0$ and $y^1 \in \Omega$. Furthermore, $v^1 \geq 0$ on $\mathbb{R}^N$.

Proof of Step 7. Because $u_n^1 \rightharpoonup v^1$, a duality argument using Corollary 2.3 shows that $v_n^1 \rightharpoonup 0$ in $\mathcal{D}^{1,p}(\mathbb{R}^N)$ if $\lambda^1 > 0$. Of course, this would contradict the previous step and so we must have $\lambda_n^1 \rightarrow 0$. Using this, it will be possible to deduce the non-negativity of v^1 . Thus, for almost every fixed $x \in \mathbb{R}^N$, one has

$$v_n^1(x) = (\lambda_n^1)^{\frac{N-p}{p}} u_n^1(\lambda_n^1 x + y_n^1) - (\lambda_n^1)^{\frac{N-p}{p}} v_0(\lambda_n^1 x + y_n^1).$$

However, for each $n \geq 1$ there holds:

$$\int_{\mathbb{R}^N} \left| (\lambda_n^1)^{\frac{N-p}{p}} v_0(\lambda_n^1 x + y_n^1) \right|^{p^*} = \int_{\mathbb{R}^N} (\lambda_n^1)^N$$

Next, arguing as in the proof of Theorem C for a suitable half-space H , one can show that v_1 is a non-trivial solution to the problem

$$\begin{cases} -\Delta_p u \equiv \mu u_+^{p^*-1} & \text{in } H, \\ u \equiv 0 & \text{on } \partial H. \end{cases}$$

In particular, we would have $u \in \mathcal{D}_0^{1,p}(H)$, thereby contradicting Theorem A. This proves the assertion. $\square$

By this last step, we know that λ_n^1 converges to 0 as $n \rightarrow \infty$, and that y_n^1 converges to a point y^1 belonging to the open set Ω . Hence, we may actually assume that $y_n^1 \in \Omega$ for each $n \in \mathbb{N}$. By virtue of the observations made above Step 5, we may assume that Ω_n converges to a half space H . Furthermore, by applying Step 1 together with Theorem C, we infer that v^1 is a critical point of Φ_∞ . From there, we find that

$$S_p \|v_1\|_{L^{p^*}(\mathbb{R}^N)}^p \leq \|v_1\| \leq \|u_+\|_{L^p(\mathbb{R}^N)}^p.$$

Thus, we must have

content...

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